

Project SG1 Design Document

CMP 4500 Intro to software profession

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Major Revisions: Changed logic that checked if a file name was valid

Pseudo Code

BEGIN

FUNCTION get_valid_filename():

 LOOP:

 PROMPT user for CSV filename

 IF filename ends with ".csv" AND file exists:

 RETURN filename

 PRINT "Error: Invalid file. Try again."

FUNCTION process_csv_file(filename):

 OPEN filename, READ data into list

 IF data has fewer than 2 rows:

 PRINT "Error: CSV file must have at least two lines."

 RETURN

 EXTRACT names from header, INIT lists: dates, presence_absence

 FOR each row (excluding header):

 APPEND first column to dates

 APPEND "1" if value > 0, else "0" to presence_absence

 WRITE names to "Names.txt"

 WRITE dates to "DatedData.txt"

 WRITE presence/absence matrix to "PresentAbsent.txt"

FUNCTION main():

 PRINT program description

 filename ← get_valid_filename()

 process_csv_file(filename)

 PROMPT "Press ENTER to exit."

IF __name__ == "__main__":

 main()

END